

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Mechatronics Engineering) SEMESTER - 5 Winter 2025 ( Regular )**

**Course :Bachelor of Technology (Mechatronics Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 5**

**Subject Code & Name: BTMXC502 - MANUFACTURING TECHNOLOGIES**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
1. What is casting in manufacturing?  
a) Molding of metal b) Cutting of metal  
c) Joining materials d) Assembling parts
  2. Pattern used in casting process is a replica of.....  
1. Final product 2. Mold 3. Runner 4. Riser
  3. Which of the following casting methods utilises wax pattern  
1. Shell moulding 2. Plaster moulding  
3. Slush casting 4. Investment casting
  4. Which of the following is also known as universal chuck?  
a) Four jaw chuck b) Three jaw chuck  
c) Both three jaw chuck and four jaw chuck d) Two jaw chuck
  5. Which of the following metal forming processes is best suitable for making the wires?  
a) Forging b) Extrusion c) Drawing d) Rolling
  6. Cold working is done at temperature.....  
a) Lower than melting point b) Higher than melting point  
c) Lower than recrystallization temperature d) Higher than recrystallization temperature
  7. Which process gives high dimensional accuracy and good surface finish.....  
a) Sand Casting b) forging c) Lapping d) Milling
  8. The material used for a forging process must be.....  
a) Brittle b) Magnetic c) Ductile d) Brittle and ductile
  9. What type of electrode is used in Submerged-arc Welding?  
a) Non-consumable b) Non-consumable and consumable  
c) Consumable d) None of the given
  10. In brazing, what temperature range is generally used?  
a) Below 450°C b) 450°C - 850°C  
c) Above 850°C d) 1000°C - 1200°C

- 11 Which material is commonly used for making lathe cutting tools?  
a) High-speed steel b) Aluminum c) Plastic d) Cast iron
- 12 What type of tool is used to create external threads?  
a) Reamer b) Taps c) End mill d) Twist Drill
- Q2. Solve the following.
- A) What do you mean by Casting Process ? Explain the steps involved in casting with 6 neat sketch?
- B) Differentiation between solidification of Pure metals and Alloys? 6
- Q3. Solve the following.
- A) Explain with neat sketch working principle of : 6  
(a) Rolling (b) Forging (c) Extrusion
- B) Explain the working of different rolling mills with neat sketch? 6
- Q4. Solve Any Two of the following.
- A) Differentiate between welding, brazing, and soldering (3 points) ? 6
- B) Explain TIG Arc welding process with neat sketch? 6
- C) Explain the principle of Resistance welding and draw sketch for spot and seam? 6
- Q5. Solve Any Two of the following.
- A) List out the different operations that can be performed on a lathe machine. Explain 6 any two with neat sketch?
- B) Draw a neat labelled diagram of Twist Drill (Drill bit) and explain its 6 nomenclature?
- C) A workpiece of diameter 150 mm and length 300 mm is to be turned at a cutting 6 speed of 100 m/min with a feed of 0.25 mm/rev. The over travel at the start and end is 5 mm each. Calculate the machining time for a single pass?
- Q6. Solve Any Two of the following.
- A) Describe the following milling processes with neat sketch: 6  
a) Peripheral milling b) Face milling
- B) What is broaching? Explain the difference between pull and push broach? 6
- C) A slab milling operation is being carried out on a 300 mm long, 100 mm wide 6 mild steel block at a feed of 0.25 mm/tooth, depth of cut 3 mm, the cutter diameter is 50 mm, and number of teeth is 20 with a cutter speed of 100 rpm. Calculate the material removal rate (MRR) in mm<sup>3</sup>/min and the cutting time in minutes. (Assume cutter width = block width).

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